RTL TOOL OUTCOMES • KEV v1.0

What can KEV tell us about an RTL design?

The tests, the completed Condor runs, and what the scores mean

Four model sizes • 151 retained designs • 725 held-out questions
15 October 2026

What the completed runs show

Different tasks reveal different strengths. A single pooled score hides important failures.

Better with 27B

Compilation errors:
8 of 9 detected

Area levels:
61 of 92 exactly right

Synthesis is weak

Synthesis failures:
4 of 24 detected

Failure reasons:
15 of 16 right when failure is supplied

No single winner

9B leads area ranking.
27B leads power ranking.

Absolute power and clock-frequency levels remain unreliable.

All four runs completed. Every model answered every test question; missing data does not explain the errors.

Compilation and synthesis answer different questions

RTL = register-transfer level: source code that describes digital hardware.

Does it compile?

Can the simulator compiler accept the source code?

Reference: QuestaSim vlog -sv.

This does not test whether the circuit is correct.

Does it synthesize?

Can Design Compiler build a complete gate-level netlist from the code?

No unresolved modules.
All registers map to cells.
No synthesis errors.

What does it cost?

Estimate cell area, power, and timing from the RTL.

The reference comes from synthesis and power tools.

No fabricated-chip or post-route measurement.

In the test set, 15 designs compile successfully but still fail synthesis.

The dataset and the held-out test set

One model request can contain several questions. Questions are not independent design samples.

151 retained designs

348 requests across both splits

875 questions across both splits

Held-out request group	Requests	Questions	What is covered
Basic checks	123	246	Compilation + synthesis
Failure reason	16	16	One of ten reason options
Performance	92	361	Area, power and two timing levels
Pair comparisons	50	102	25 distinct pairs, both orders
TOTAL TEST	281	725	123 individual test designs

Development: 28 designs, 67 requests, 150 questions. Related base designs stay in the same split.

What the model receives for each test

Tool logs and correct answers are used for scoring; they are not sent to the model.

Basic checks

RTL source + top module + tool description.
Questions: does it compile, and does it synthesize?

Failure reason

The same source, plus a prompt saying synthesis failed.
Choose one reason from ten descriptions.

Performance

RTL + flow + clock target; power also gets its testbench.
Choose a level for each available metric.

Pair comparison

Both RTL variants + the same flow and clock.
Power comparisons also share the same testbench.

The failure-reason task supplies a key fact that the synthesis yes/no task must discover.

How to read the scores

Every result needs a denominator and a clear rule for what counts as correct.

Yes or no

A yes probability of 0.5 or more means yes.

Balanced accuracy gives “yes” and “no” designs equal weight.

Always one answer: 50%.

Five levels

Exact: the chosen level matches the tool label.

Within one: a difference of at most one level.

This does not mean a fixed physical error tolerance.

Choose A or B

Each pair is sent twice, with its order swapped.

We report per-view accuracy and check both orders.

50% is an always-A or always-B reference here.

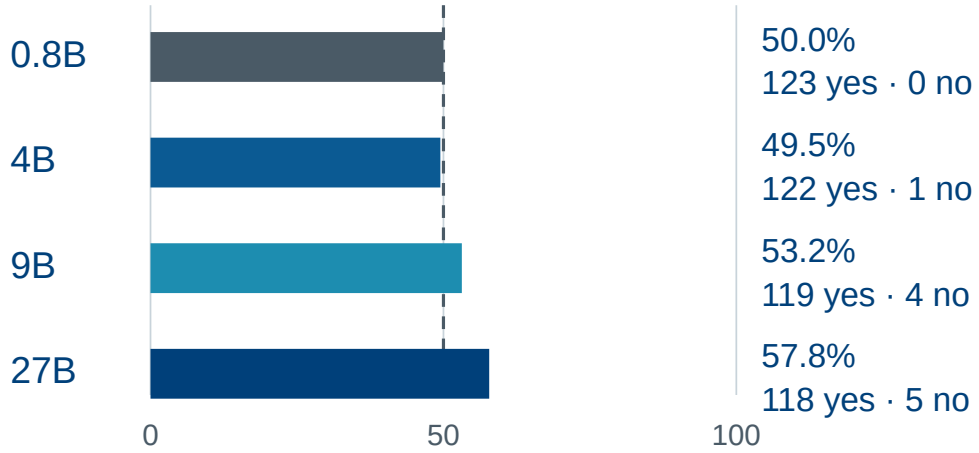
A “constant-answer reference” below means always choosing one stated answer, calculated from test labels.

Only KEV-27B’s compilation score is far above 50%

One score per model: “yes” and “no” designs get equal weight. A constant answer scores 50%.

Does it synthesize?

Tool answers for the 123 designs: 99 yes · 24 no

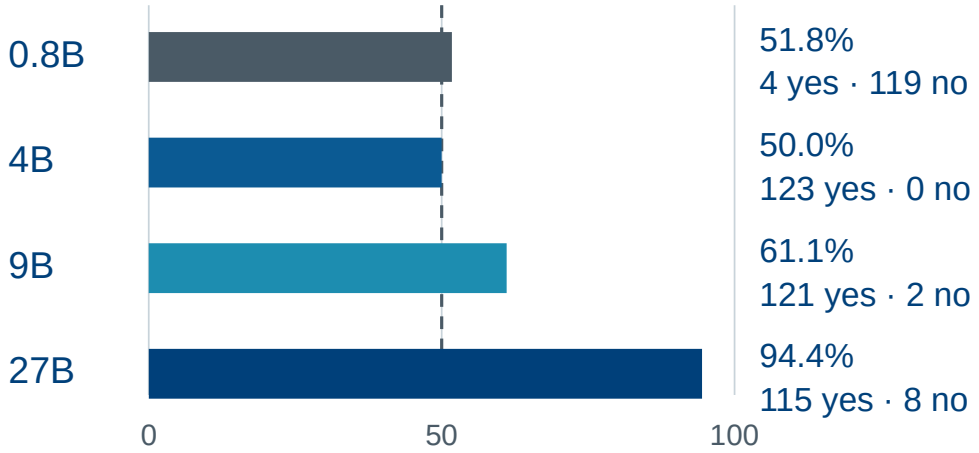


Balanced accuracy (%) — higher is better

Dashed line: always “yes” or always “no” = 50%

Does it compile?

Tool answers for the 123 designs: 114 yes · 9 no



Balanced accuracy (%) — higher is better

Dashed line: always “yes” or always “no” = 50%

In the other seven cases, at least 118 of the 123 answers are the same.

Why 82.9% hid poor synthesis failure detection

KEV-27B • 123 held-out designs • a yes/no threshold of 0.5

Actual tool result	Model says YES	Model says NO
Synthesizes — 99 designs	98 correct	1 wrong
Fails synthesis — 24 designs	20 wrong	4 correct

20 failures missed

82.9% overall accuracy

102 correct of 123 designs.
Each design has equal weight; the 99 successes dominate.

57.8% balanced accuracy

Average of 99.0% success recognition and 16.7% failure detection. Each group gets half the weight.

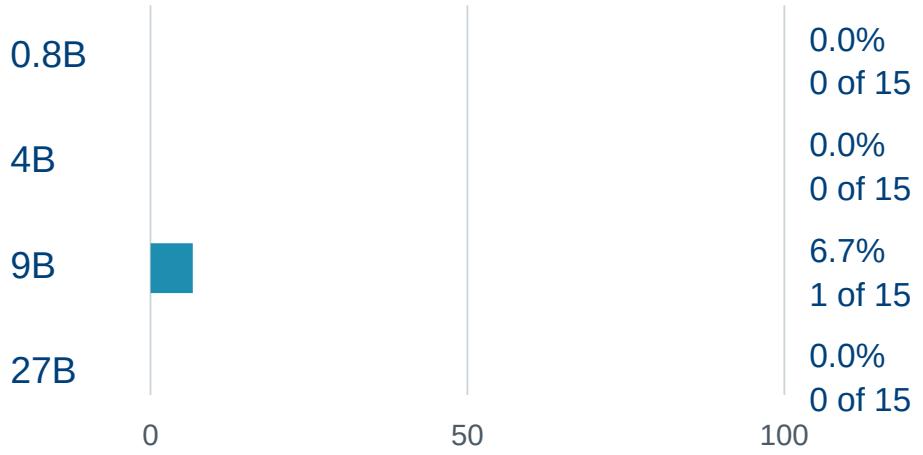
Both percentages are correct. They answer different questions about an imbalanced test set.

Which synthesis failures do the models detect?

Split the 24 failures again: source accepted by the simulator compiler versus rejected source.

Compiles, but fails synthesis

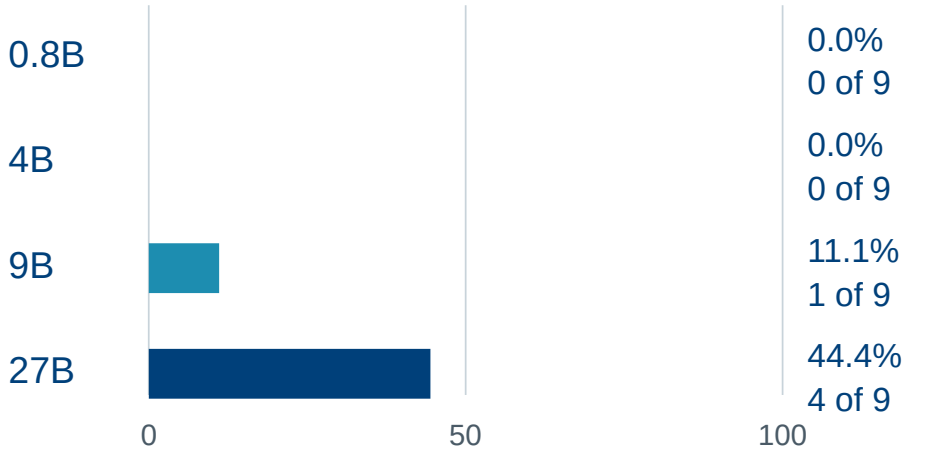
15 designs • synthesis-specific failure cases



Correct answers (%) — higher is better

Compilation also fails

9 designs • both tools reject the design



Correct answers (%) — higher is better

KEV-27B misses every one of the 15 synthesis failures whose source compiles.

Failure reason: choose one of ten options

The prompt says that synthesis failed. 25 reason questions: 16 test, 9 development (dev).

Option	Text of the option, as the model reads it	Test	Dev
illegal_code	The code is not legal Verilog or SystemVerilog: a syntax error, an undeclared name, a non-constant range, or a procedural assignment to a net.	1	3
unsupported_construct	The code is legal for a simulator, but it uses a data type or construct that synthesis does not support: real, time, string, trireg, a class, a dynamic array, a DPI import, or a user-defined primitive.	1	0
timing_control	The code uses a timing control that synthesis does not support: wait, fork-join, or an event trigger (->).	3	1
simulation_statement	The code uses a statement or system function for simulation only: force-release, procedural assign-deassign, or $random.	5	1
event_list	An event list that synthesis does not support: both edges of one signal, edge and level events mixed, or different edges in one process.	0	1
async_reset	An asynchronous reset or set that synthesis cannot build: statements outside the reset if-else, a second edge signal without the reset template, or a reset to a non-constant value.	1	0
multiple_drivers	A signal has more than one driver, for example assignments in two always blocks.	1	2
mixed_assignments	One variable gets blocking and nonblocking assignments in the same always block.	2	0
unbounded_loop	A loop or a recursion that synthesis cannot unroll: a forever loop with no timing control, or a recursion with no constant limit.	0	1
instance_mismatch	A module instance does not match a module: a missing module, an unknown parameter, or an unknown port.	2	0

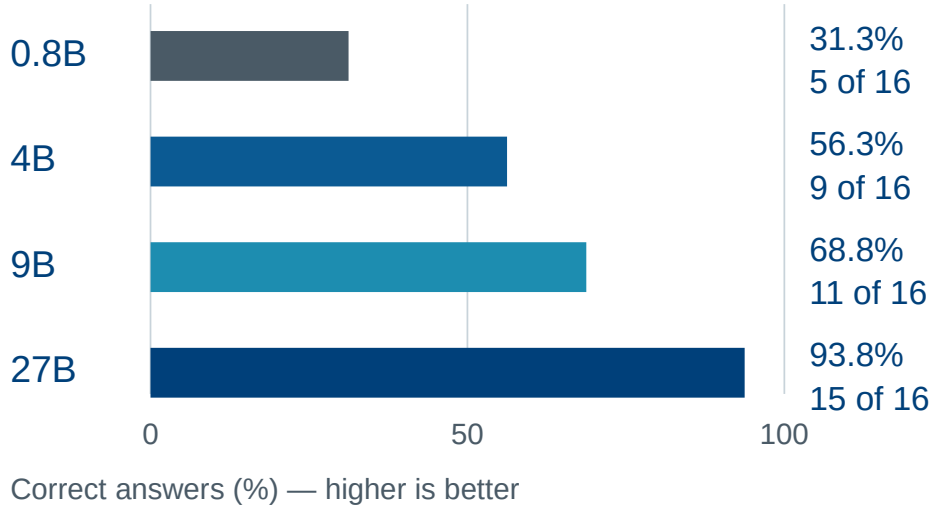
22 of the 25 failures are injected edits. 8 more test designs with a compile error get no reason question.

Failure reasons: 27B gets 15 of 16 right

The prompt explicitly states that synthesis failed, then offers ten possible reasons.

Choose the correct failure reason

16 known-failure designs • one choice per design



What is measured

Can the model identify the reason when failure is already known?

What the 27B error was

A multiple-driver error was labelled as an asynchronous-reset error.

Only eight of the ten reason classes occur in these 16 test designs. Coverage is limited.

Diagnosing a known failure is not detecting that failure

Match the two saved responses for the same 16 reason-test designs.

Model	Says synthesis failed	Correct reason when told it failed	Both answers correct
KEV-0.8B	0 of 16	5 of 16	0 of 16
KEV-4B	0 of 16	9 of 16	0 of 16
KEV-9B	1 of 16	11 of 16	1 of 16
KEV-27B	0 of 16	15 of 16	0 of 16

27B: 15 correct diagnoses, but zero failure flags on these same designs.

This is a linked analysis of existing answers, not a separately run detection-and-diagnosis system.

Area and power: choose one of five ranges

The model estimates a level from source code and flow information, not an exact physical value.

Metric	Level 1	Level 2	Level 3	Level 4	Level 5
Cell area (gate equivalents)	<100	100 to <1,000	1,000 to <10,000	10,000 to <100,000	≥100,000
Total power (microwatts)	<10	10 to <100	100 to <1,000	1,000 to <10,000	≥10,000

Area reference

1 gate equivalent = one two-input NAND gate
= 0.196608 square micrometres in this library.

Power reference

Dynamic + leakage, for the supplied stimulus.
1,000 microwatts = 1 milliwatt.

Exact level is strict; “within one level” can span a large physical range because these bins are logarithmic.

Timing: margin at 500 MHz, plus a derived frequency

The clock target is 500 MHz: one cycle lasts 2 nanoseconds (ns).

Metric	Level 1	Level 2	Level 3	Level 4	Level 5
Worst slack (timing margin)	Negative: misses target	0 to <0.5 ns	0.5 to <1.0 ns	1.0 to <1.5 ns	≥1.5 ns
Reachable clock frequency	<1 GHz	1 to <2 GHz	2 to <3 GHz	3 to <4 GHz	≥4 GHz

Reachable frequency (GHz) = 1 ÷ [2 ns − worst slack (ns)]

Example: 1.438 ns of slack gives a 0.562 ns limiting delay, or about 1.78 GHz.
This is derived from the netlist synthesized at 500 MHz; it is not a new optimized run.

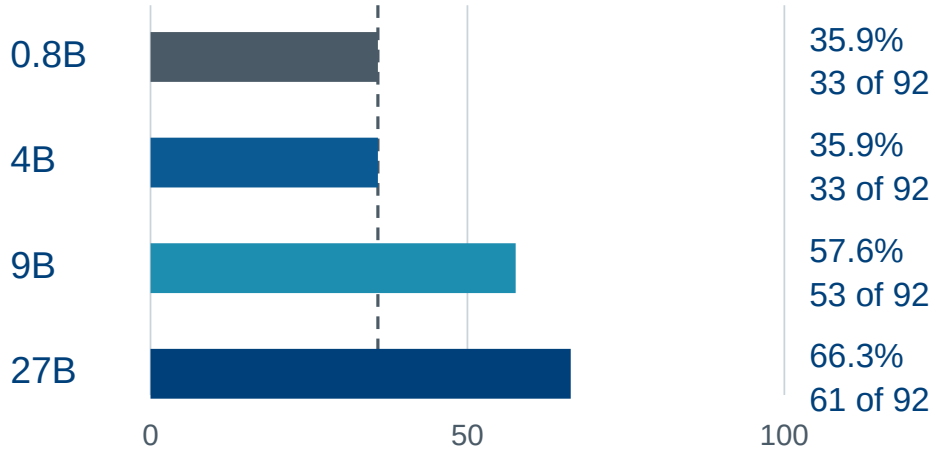
No negative-slack designs occur in this test split. Timing-violation detection is therefore not evaluated.

Area levels improve substantially with 9B and 27B

92 held-out designs • same flow and target • exact and within-one scores shown separately.

Exact area level

All five true area levels occur

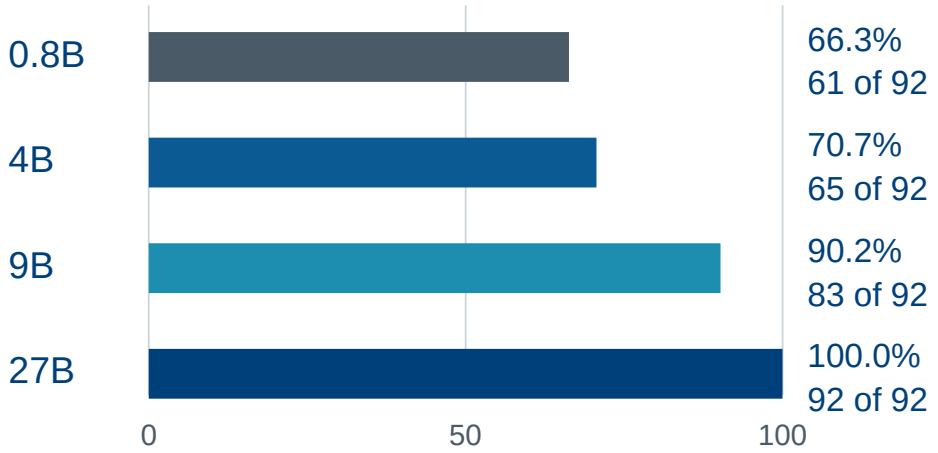


Correct answers (%) — higher is better

Dashed line: always level 1 = 35.9%

Within one area level

Chosen level differs by at most one



Correct answers (%) — higher is better

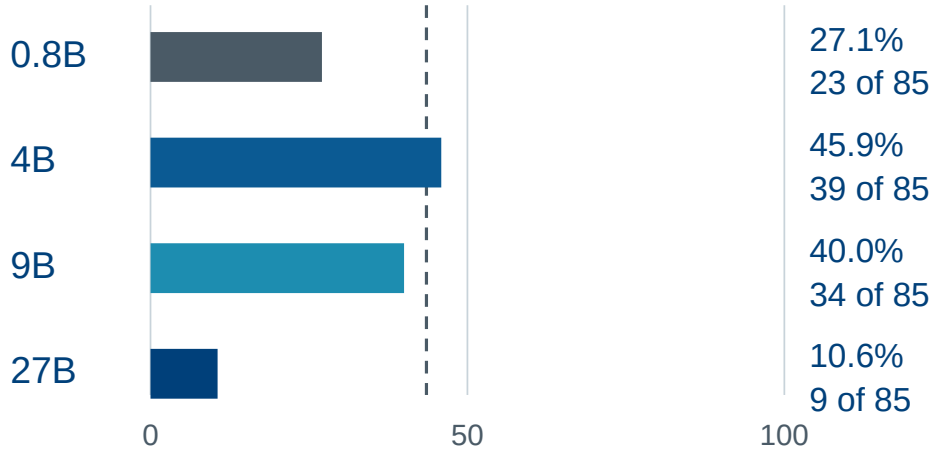
27B: 61 of 92 exact levels; all 92 estimates are within one level.

Power levels: the largest model performs worst

85 held-out designs with usable power measurements and a supplied stimulus testbench.

Exact power level

Levels 1 to 4 occur; level 5 is absent

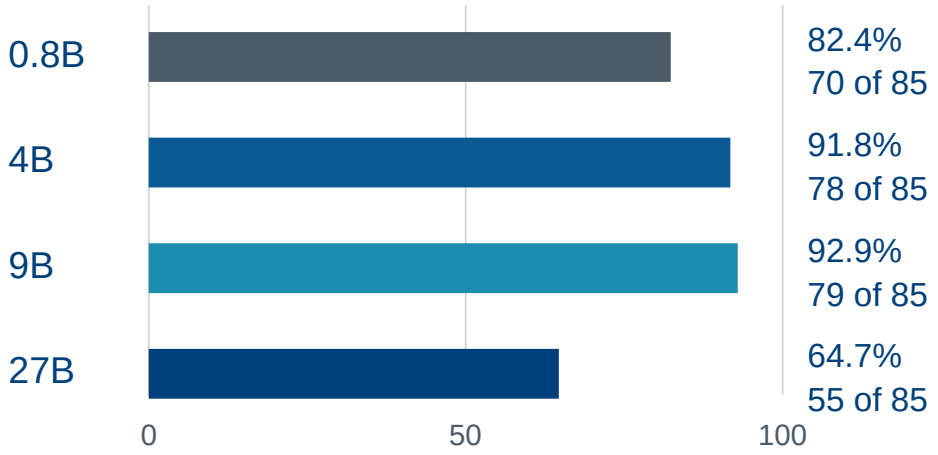


Correct answers (%) — higher is better

Dashed line: always level 2 = 43.5%

Within one power level

Chosen level differs by at most one



Correct answers (%) — higher is better

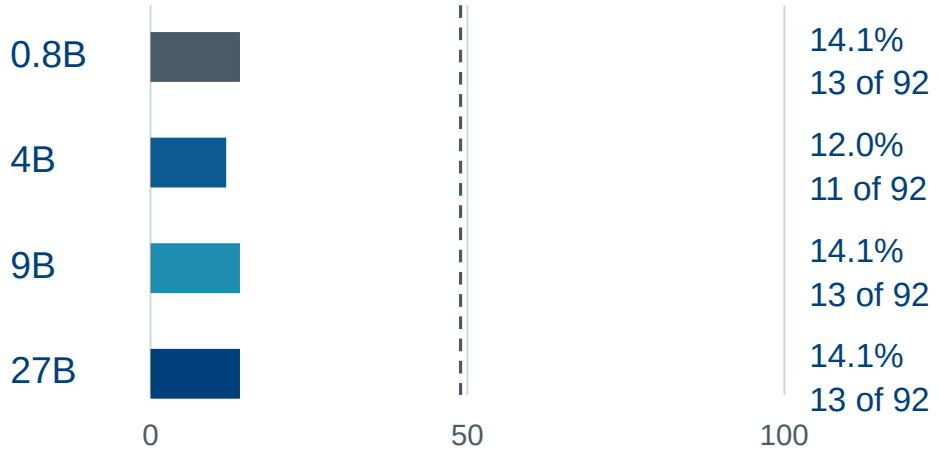
27B predicts only levels 3–5, although 60 of the 85 true labels are levels 1–2.

Timing levels remain unreliable for every model

92 held-out designs • these are level estimates from RTL, not timing-tool replacements.

Exact clock-frequency level

Frequency derived from the 500 MHz netlist

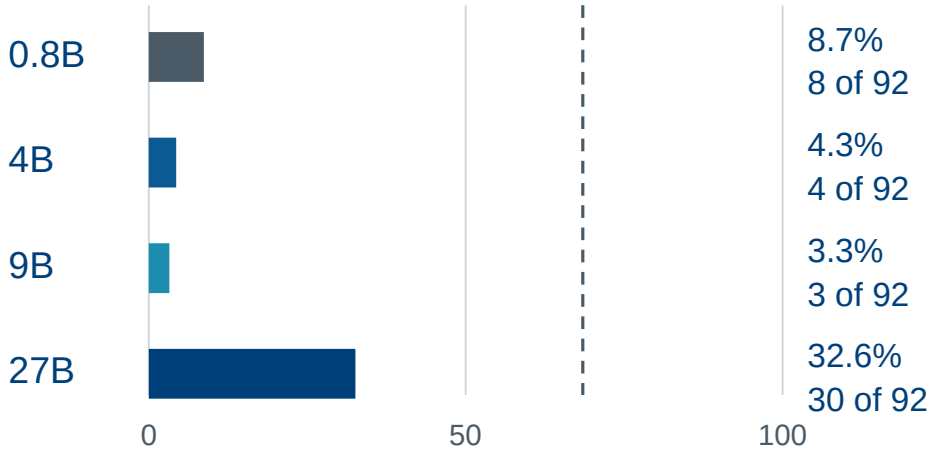


Correct answers (%) — higher is better

Dashed line: always level 4 = 48.9%

Exact timing-margin level

Worst slack at the 500 MHz target



Correct answers (%) — higher is better

Dashed line: always level 5 = 68.5%

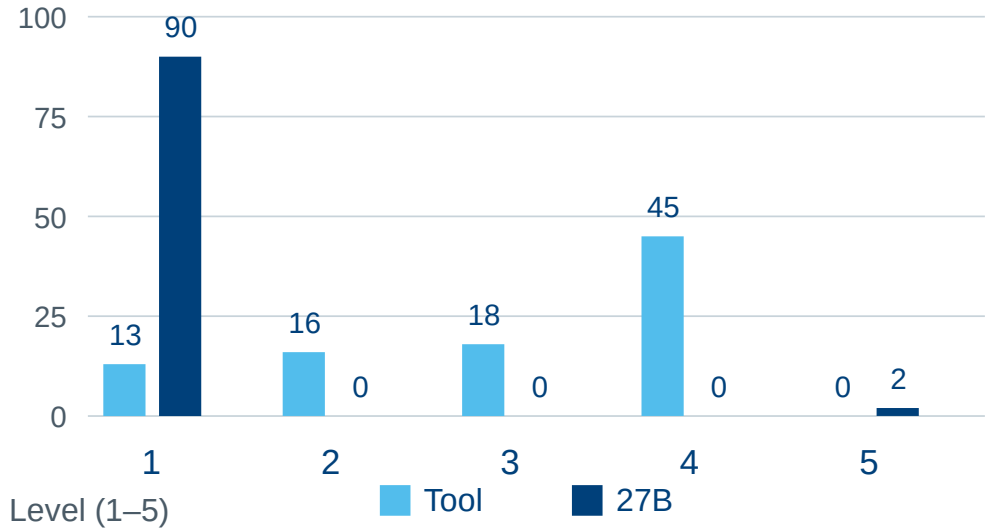
27B’s better slack result still falls below the stated constant-answer reference.

27B’s predicted levels reveal the mismatch

The chart heights are design counts, not accuracy. Every level is shown, including zero counts.

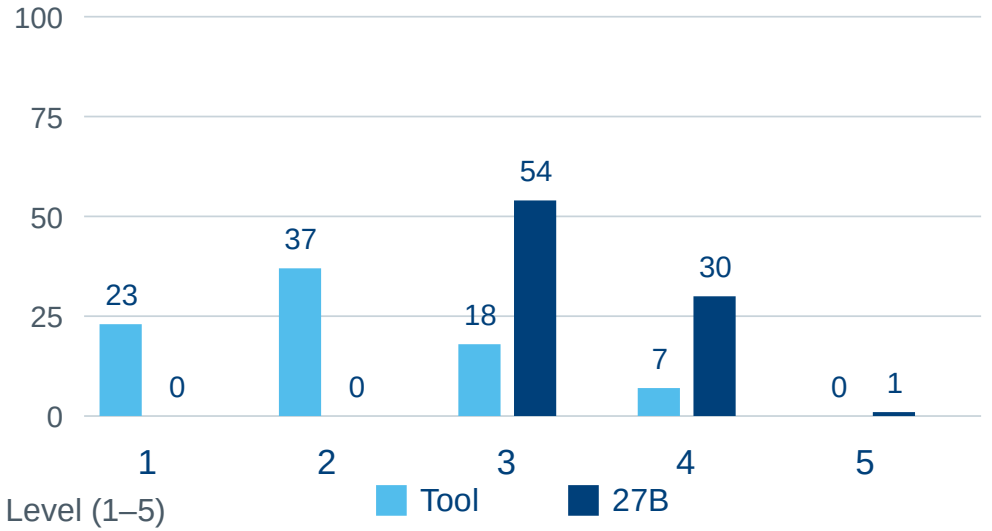
Clock-frequency levels

92 designs • counts at each level



Power levels

85 designs • counts at each level



27B usually chooses the lowest frequency level and high power levels; the tool labels have a different distribution.

Pair tests ask which variant is better

Two candidates for the same problem, with the same flow and power stimulus.

Three comparisons

Smaller cell area
Lower total power
Higher derived clock frequency

Eligibility and coverage

Metric differs by at least 10%.
Area: 23 distinct pairs, 46 views.
Power: 22 pairs, 44 views.
Frequency: 6 pairs, 12 views.

First request: [design X = A] [design Y = B]

Swap request: [design Y = A] [design X = B]

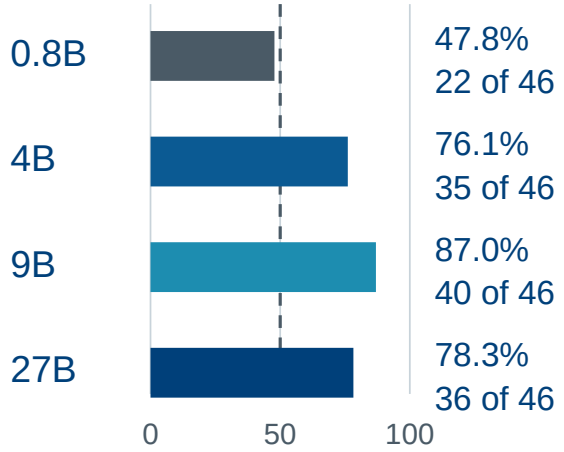
A model tracking the better design should switch letters when the design order is swapped.

Pair rankings improve, but the strongest model varies

Per-view accuracy: each distinct pair contributes its two presentation orders.

Choose the smaller area

23 distinct pairs • 46 views

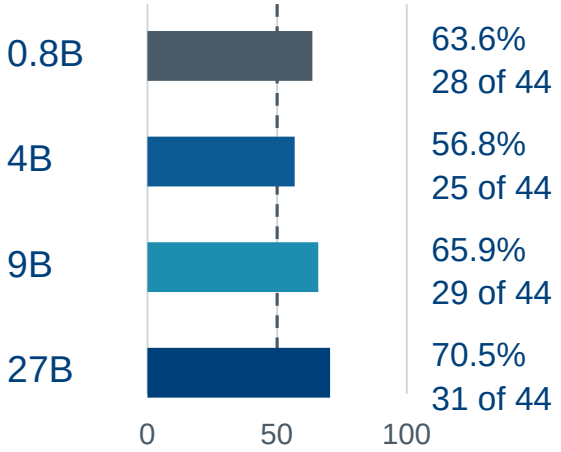


Correct answers (%) — higher is better

Dashed line: always A or B = 50%

Choose lower power

22 distinct pairs • 44 views

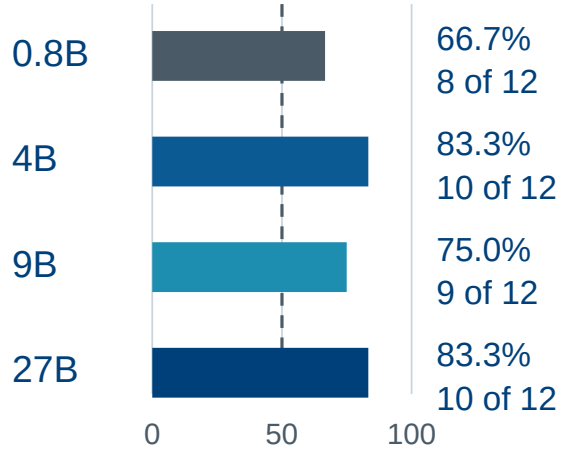


Correct answers (%) — higher is better

Dashed line: always A or B = 50%

Choose higher frequency

6 distinct pairs • 12 views



Correct answers (%) — higher is better

Dashed line: always A or B = 50%

9B leads area ranking (40 of 46 views); 27B leads power ranking (31 of 44). Frequency has only six pairs.

Check the design ranking in both orders

A stronger test counts a pair correct only when both presentation orders are correct.

Model	Area: both correct 23 pairs	Power: both correct 22 pairs	Frequency: both correct 6 pairs
KEV-0.8B	5 of 23	6 of 22	3 of 6
KEV-4B	13 of 23	3 of 22	4 of 6
KEV-9B	19 of 23	9 of 22	4 of 6
KEV-27B	17 of 23	14 of 22	5 of 6

Same letter after swapping the power pair:

0.8B: 16 of 22 pairs • 4B: 19 of 22 • 9B: 11 of 22 • 27B: 3 of 22

Keeping the same letter after a swap guarantees at least one wrong view. Lower is better for that error count.

All four Condor evaluations completed

Observed run times include different hardware and setup costs; they are not a controlled speed ranking.

Model	Job	GPU used	Job time including setup	Median model time per request
KEV-0.8B	4.0	RTX 3080 Ti	125 s	109 ms
KEV-4B	5.0	A100 80 GB	187 s	201 ms
KEV-9B	6.0	A100 80 GB	204 s	244 ms
KEV-27B	7.0	A100 80 GB	400 s	668 ms

Every model: 348 dataset requests answered, 5 smoke requests passed, 725 test questions scored, zero HTTP errors and zero missing answers.

All runs used the same dataset fingerprint and v1.0 checkpoints. Maximum reported input length: 7,706 tokens.

What the evidence supports—and what needs work

These are observations on dataset v1, not a guarantee for unseen design families.

Promising in this test

27B: source-compilation screening.
27B: failure reasons when failure is known.
27B: coarse area levels.
9B: pairwise area ranking.
27B: pairwise power ranking.

Next experiments

Choose binary thresholds on development data.
Expand synthesis failures and rare classes.
Add designs that miss the timing target.
Check power and timing across design groups.
Compare other models under the same protocol.

Keep tool verification in the loop. Reliable synthesis-failure screening is not established by these scores.

THE DETAIL BEHIND THE NUMBERS

Full results and coverage limits

Exact counts, probability metrics, level errors and reproducible evidence

Every primary test score

Correct answers (%) • each question has equal weight within its row • higher is better.

Task	Questions	0.8B	4B	9B	27B
Compiles — pooled	123	10.6%	92.7%	94.3%	99.2%
Synthesizes — pooled	123	80.5%	79.7%	80.5%	82.9%
Failure reason — known failure	16	31.3%	56.3%	68.8%	93.8%
Area — exact level	92	35.9%	35.9%	57.6%	66.3%
Power — exact level	85	27.1%	45.9%	40.0%	10.6%
Clock frequency — exact level	92	14.1%	12.0%	14.1%	14.1%
Timing margin — exact level	92	8.7%	4.3%	3.3%	32.6%
Pair: smaller area	46	47.8%	76.1%	87.0%	78.3%
Pair: lower power	44	63.6%	56.8%	65.9%	70.5%
Pair: higher frequency	12	66.7%	83.3%	75.0%	83.3%

Pooled yes/no scores conceal class imbalance. Use the successful/failing split on the next slide.

Yes/no checks: every class, count and weighting

“Correct” below is measured within the actual tool-outcome group.

Actual tool group	Designs	0.8B	4B	9B	27B
Compilation: successes	114	4 (3.5%)	114 (100.0%)	114 (100.0%)	114 (100.0%)
Compilation: failures	9	9 (100.0%)	0 (0.0%)	2 (22.2%)	8 (88.9%)
Compilation: balanced	—	51.8%	50.0%	61.1%	94.4%
Compilation: “yes” answers	123	4	123	121	115
Synthesis: successes	99	99 (100.0%)	98 (99.0%)	97 (98.0%)	98 (99.0%)
Synthesis: failures	24	0 (0.0%)	0 (0.0%)	2 (8.3%)	4 (16.7%)
Synthesis: balanced	—	50.0%	49.5%	53.2%	57.8%
Synthesis: “yes” answers	123	123	122	119	118

Group rows: number correct, then the percentage of that group. “Yes” rows count all 123 designs.
Balanced accuracy averages success recognition and failure detection with 50% weight each.

Most failing designs get the same answer as successes

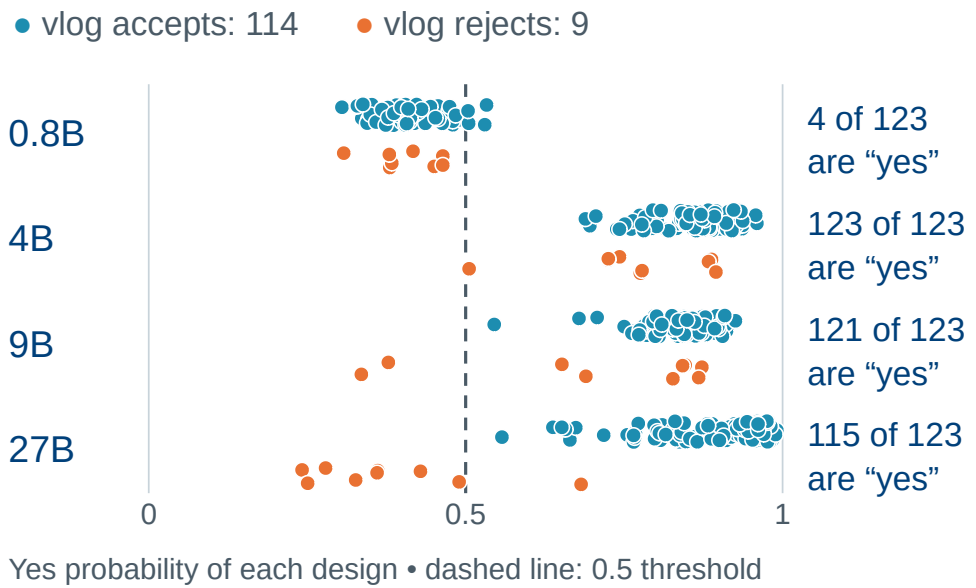
Each dot is one held-out design. A yes probability of 0.5 or more counts as “yes”.

Does it synthesize?



Yes probability of each design • dashed line: 0.5 threshold

Does it compile?



Yes probability of each design • dashed line: 0.5 threshold

Only KEV-27B’s compile answers are right in both groups: 114 of 114 and 8 of 9.

Failure-reason coverage is small and uneven

Correct reason counts by actual class • the prompt supplies that failure occurred.

Actual failure class	Cases	0.8B	4B	9B	27B
More than one driver	1	0	0	0	0
Illegal code	1	0	0	0	1
Unsupported construct	1	0	1	1	1
Unsupported timing control	3	1	3	3	3
Simulation-only statement	5	3	5	5	5
Asynchronous reset	1	0	0	0	1
Mixed assignment styles	2	0	0	0	2
Module-instance mismatch	2	1	0	2	2
Equal weight per observed class	Accuracy	17.9%	37.5%	50.0%	87.5%

No true event-list or unbounded-loop cases occur in this held-out subset.
These two offered classes therefore have no measured class recall.

Probability ranking and binary decisions are different

A model can rank cases reasonably while making poor decisions at the fixed 0.5 threshold.

Probability metric	0.8B	4B	9B	27B
Compilation: ranking score (AUROC)	0.563	0.763	0.692	0.994
Compilation: probability error (Brier)	0.334	0.064	0.066	0.030
Synthesis: ranking score (AUROC)	0.490	0.714	0.740	0.814
Synthesis: probability error (Brier)	0.187	0.148	0.144	0.128

Ranking score: higher is better

Chance = 0.5; perfect ordering = 1.
Compares yes probabilities for successes and failures.

Probability error: lower is better

Mean squared error of the yes probability.
Zero is perfect; confidently wrong answers cost more.

Performance levels: tolerance and expected-level error

Within one level: higher is better. Mean level error: lower is better.

Metric	Measure	0.8B	4B	9B	27B
Area	Within one level	66.3%	70.7%	90.2%	100.0%
Area	Mean level error	0.994	0.713	0.649	0.485
Power	Within one level	82.4%	91.8%	92.9%	64.7%
Power	Mean level error	0.971	0.754	0.900	1.070
Clock frequency	Within one level	31.5%	31.5%	33.7%	31.5%
Clock frequency	Mean level error	1.119	1.321	1.208	1.242
Timing margin	Within one level	30.4%	28.3%	12.0%	40.2%
Timing margin	Mean level error	1.509	1.654	1.742	1.622

Mean level error uses the probability-weighted expected level, not just the most likely level.
Unequal physical bin widths mean that one level is not a fixed physical error.

Pair order: all correctness and letter-bias counts

Counts are per distinct pair. Each pair has exactly two model answers.

Pair outcome	Pairs	0.8B	4B	9B	27B
Area — Both orders correct	23	5	13	19	17
Area — One order correct	23	12	9	2	2
Area — Neither order correct	23	6	1	2	4
Power — Both orders correct	22	6	3	9	14
Power — One order correct	22	16	19	11	3
Power — Neither order correct	22	0	0	2	5
Frequency — Both orders correct	6	3	4	4	5
Frequency — One order correct	6	2	2	1	0
Frequency — Neither order correct	6	1	0	1	1

With two opposite correct letters, “one order correct” is the same as keeping the same letter.
A consistent letter swap can still be wrong in both orders.

Dataset composition and missing coverage

The retained designs cover several sources; this is not a balanced population sample.

Design group	All retained	Held-out designs	Held-out synthesis failures
MetRex: varied sizes	40	31	0
Injected synthesis errors	16	9	9
Natural language-model candidates	20	17	0
Injected compilation errors	8	8	8
Human reference designs	15	11	0
Older REvolution candidates	32	28	1
REvolution v0 designs	10	9	0
Accepted edits and rejected twins	10	10	6

Not measured here: energy per operation, functional equivalence, post-route timing, measured chips, other model families, or the older plan’s rename/comment controls.

Evidence needed to reproduce this deck

Every number is tied to the frozen dataset and saved responses; no model reruns were needed.

Model	Condor job	Completed on 15 October	Checkpoint revision (prefix)
KEV-0.8B	4.0	11:30:55	bf75a6a8848e
KEV-4B	5.0	14:31:45	6cfce5c2fa4b
KEV-9B	6.0	14:34:55	db029f08b290
KEV-27B	7.0	14:39:12	af0e6d551bdc

Inputs and evidence

Questions: data/questions/dataset_v1.jsonl
Answers, scores and manifests: results/kev/
Verified tables and rebuild scripts: presentations/data/ and presentations/src/